Cao **Wu**

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Education

Politecnico di Milano *Milan, Italy*
M.Sc. in High Performance Computing Engineering *Sep. 2024 - Dec. 2026*
- Core coursework: Parallel Computing, Partial Differential Equations, Advanced Computer Architecture, Machine Learning, and Distributed Systems.

Sichuan University *Chengdu, China*
B.Eng. in Computer Science and Technology *Sep. 2018 - Jun. 2022*

Work Experience

Bilibili *Shanghai, China*
Golang Backend Software Development Engineer **(FULL-TIME)** *Jul. 2022 - Aug. 2024*
- Built core settlement services for Bilibili's live-streaming business with Go, MySQL, Redis, Kafka, Hive, and ClickHouse.
- Promoted within the first year and became a main developer of the anchor settlement system.
- Supported settlement flows with cumulative processed rewards exceeding RMB 1.3 billion from Jul. 2022 to Aug. 2024.

Projects

GrowYumi *Personal WeChat Mini Program*
Founder *Feb. 2022 - Present*
- Built and operated a personal WeChat Mini Program for couples, reaching **100,000+** registered users.
- With access to the WeChat advertising platform, monthly income can reach RMB 1,000 during peak periods.

A Simplified vLLM Core *LLM Inference Systems*
Personal Project *Dec. 2025 - Present*
- Built a simplified vLLM-style inference core from scratch in PyTorch/CUDA, implementing the Engine, Scheduler, and BlockManager to connect request intake, scheduling, execution, and response generation end to end.
- Implemented paged KV cache, continuous batching, and mixed prefill/decode scheduling to support concurrent offline inference for multiple requests while improving memory efficiency and batch throughput.
- Added AWQ quantization to reduce model weight storage overhead, and structured the system around reusable cache-management and execution abstractions to support future sampling, parallelism, and performance optimizations.

RGB-Infrared Fusion for Vehicle-Scene Segmentation *Sichuan University*
Graduation Project *Sep. 2020 - Nov. 2021*
- Improved semantic segmentation under low-light and glare conditions with RGB-infrared fusion.
- Used PyTorch and a temporal-difference method to improve recognition efficiency.

Selected Awards

2021	**Member of ACM Algorithm Training Team (One of 32 members)**, Sichuan University
Apr. 2021	**Second Prize, Southwest Division (Top 15%)**, China University Computer Contest
Oct. 2021	**Provincial Second Prize (Top 20%)**, Lanqiao Cup C/C++ Algorithm Programming Competition
Oct. 2020	**Silver Medal (Top 10%)**, China Collegiate Algorithm Design & Programming Challenge Contest

Skills

Languages & Frameworks	Python, C++, CUDA, C, Go, PyTorch, Hugging Face
LLM Inference	KV cache, PagedAttention, continuous batching, vLLM, Transformers, Qwen
CUDA Kernels	Matrix multiplication, transpose, reduction, FlashAttention-style prefill, GPU memory optimization